Report C — Project Working Report

Every tiny step from loading a file to the YES / NO tumor-candidate decision, with sample outputs.

This report walks the software the way a beginner should learn it: one module at a time, and inside each module one small action at a time. For every action we say the **input**, the **technique**, and the **output**. Formulas are the ones in the code, not textbook-only versions. Pictures come from the in-repo sample files.

Important: the reconstructed image and the YES/NO label are research estimates. They are not a medical diagnosis.

How to read this report

- The project has three teaching modules: (1) load data, (2) clean the signal, (3) make an image and find a region.
- Automation later repeats the same chain on many files. Auto Tweak only searches geometry on one already-loaded scan.
- sample_touchstone.s2p is used for Module 1–2 pictures. Reconstruction pictures use the 8-antenna synthetic family of sample_matlab.mat, because a single S21 trace cannot form a useful multi-antenna image.
- When a number is compared to a limit, that limit is written next to the step.

0. The whole workflow in one place

```
Load file ↓ Module 1 MicrowaveDataset (frequencies + complex S) ↓ Module 2 Cleaned traces (or unchanged BMID _adi data) ↓ IFFT Time-domain traces ↓ DAS / DMAS / DMAS-D4 Three images ↓ beamformer selection One selected image ↓ ROI detection One or more boxes ↓ high-resolution rebuild of the box Refined local image ↓ characterization + suspicion YES / NO tumor-candidate + confidence
```

The purpose changes at each arrow. First we only want a valid matrix. Then we want a cleaner matrix. Then we want a picture. Then we stop asking “which algorithm looks nicest?” and start asking “where is the most relevant region in the chosen picture?” Then we describe that region with numbers. Only after that do we decide whether it looks tumor-like under the project’s rules.

Module 1 — Data acquisition (loading)

Module 1 does not make an image. It only turns a file on disk into one common object called MicrowaveDataset.

Step 1. The user (or automation) chooses a path

Input: a file path such as datasets/sample_touchstone.s2p or a UM-BMID fd_data_*.mat plus a scan index.
Technique: look at the extension. .mat goes to the MATLAB loader. .s1p/.s2p/.s4p/.s8p go to the Touchstone loader. Anything else raises UnsupportedFileFormatError.
Output: a decision about which parser to run. Nothing is reconstructed yet.

Step 2. If the file is Touchstone .s2p

The parser reads comment lines (starting with !) and the option line (starting with #). Our sample option line is:

```
# GHz S RI R 50.0
```

That sentence means: frequencies are in GHz, the numbers are S-parameters, the pair format is Real/Imaginary, and the reference impedance is 50 ohm. Then each data row is one frequency followed by eight numbers: Re(S11) Im(S11) Re(S21) Im(S21) Re(S12) Im(S12) Re(S22) Im(S22).

```
S = Re + j Im
|S| = sqrt(Re² + Im²)
phase = atan2(Im, Re)
|S|_dB = 20 log10(|S|)
If the file had been DB format: |S| = 10^(dB/20), then S = |S|(cos θ + j sin θ).
```

Input: the text file.
Technique: scikit-rf Network plus a project S21 helper that rebuilds S21 from the eight-number rows.

Output: frequencies in Hz (length 201), S-matrix shape (n_freq, 2, 2), and metadata. Δf = 40.00 MHz. First S21 = +0.126712 -0.018170j.

Step 3. If the file is a general MATLAB .mat

The loader first asks: is this HDF5 (MATLAB v7.3) or a legacy MAT? Then it lists numeric variables. It looks for a frequency vector by name (freq, frequency, …) or by "real, increasing, looks like RF". If the largest frequency is less than 1000, it treats the vector as GHz and multiplies by 1e9. It then picks the S array (name hints such as s_param / s21, plus "has the frequency length in its shape"). Axis 0 is forced to be frequency. A 2-D array stays (n_freq, n_traces). A square 3-D array stays (n_freq, n_ports, n_ports).

Input: .mat variables.
Technique: name heuristics + documented fallbacks (fd_data_s21 → 1–8 GHz, 1001 points).
Output: MicrowaveDataset. If no numbers exist, it tells you the file is probably metadata-only.

Step 4. If the file is a UM-BMID cube

A BMID measurement file is a 3-D complex cube. The code finds an array that contains the length 1001 and reorders it to:

```
cube[scan, frequency, antenna] ∈ complex
one scan S = cube[k, :, :] shape (1001, n_antennas)
f = linspace(1 GHz, 8 GHz, 1001)
```

If you do not give scan_index, loading stops with ScanSelectionRequiredError. That is on purpose: the cube is many experiments in one file. Companion md_list_*.mat supplies labels: tum_diam or tum_rad, tum_x, tum_y, birads, tum_shape, ant_rad, phant_id. A scan is called tumor if diameter > 0 or a non-zero (x, y) exists. Radius 18 cm and FOV ±6 cm are stored in metadata.

Input: fd_data_*.mat + optional md_list_*.mat + scan index.
Technique: slice one scan; attach labels in metres.
Output: one MicrowaveDataset for that scan only.

Step 5. Validation before anyone plots

Frequencies and S-values must be finite. Empty files fail. Wrong Touchstone headers fail. The GUI shows the summary panel: file name, type, number of samples, number of frequencies, range, ports, size, variable names. Number of samples means how many complex entries are stored, not how many tumors exist.

Sample Module 1 output — sample_touchstone.s2p

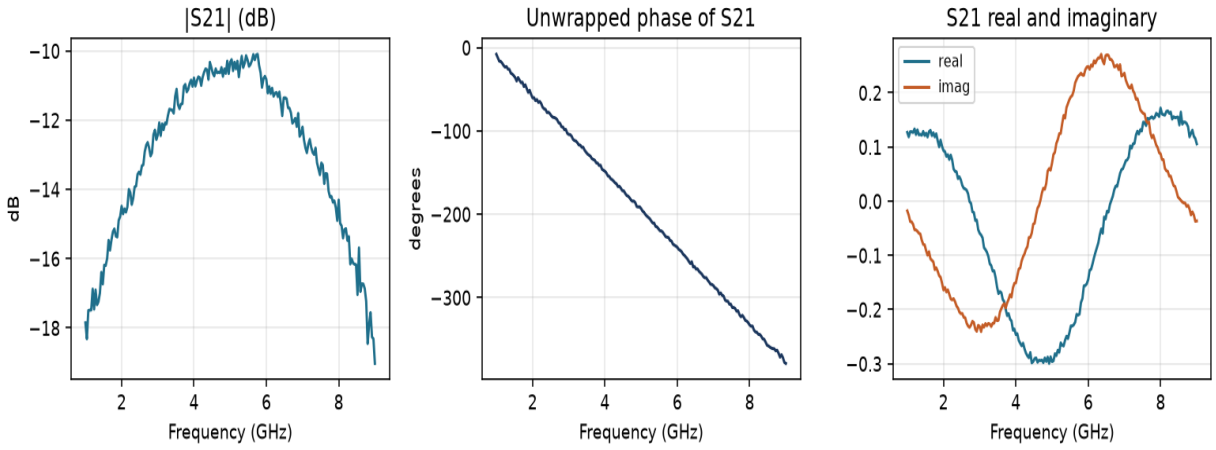


Figure. Left: |S21| in dB across 1–9 GHz. Middle: unwrapped phase (the delay story). Right: the actual real and imaginary parts that later stages filter. At 1 GHz, |S21| ≈ -17.86 dB. This is a synthetic teaching file, so the curve is a smooth resonance plus noise, not a hospital measurement.

How to interpret this figure. A falling then rising |S21| means the two-port sample has a frequency-dependent transmission. A steadily changing unwrapped phase means delay. If real and imaginary were both exactly zero, the file would be empty of energy and later reconstruction would fail the energy gate. Module 1's job ends when these arrays sit in memory. It does not decide if a tumor exists.

Module 2 — Signal preprocessing

The purpose now changes. We already have a valid matrix. We ask: can we make the traces cleaner without inventing a fake target? There are two tracks.

Track	When	What happens
.s2p UG track	file ends with .s2p	Extract S21 only, then Week 1–3 cleaning
General MATLAB	other .mat / multi-trace	filter → calibrate → normalize → background → artifacts
BMID pass-through	dataset_family = UM-BMID	every method = none, Week 3 off

Step 6. Router

Input: MicrowaveDataset + PreprocessingConfig (GUI preset or automation defaults).
Technique: if file_type starts with Touchstone and path ends with .s2p, call process_touchstone_s21_dataset. Else run the five-stage general loop.
Output: PreprocessingResult: original dataset, processed dataset, stage snapshots, quality report, optional Week 3 object.

Step 7. Extract S21 only (.s2p track)

A 2×2 matrix has four stories. The UG brief uses transmission S21. After this step the working array is 1-D (or n_freq × 1), not 2×2. S11, S12, S22 stay in the original dataset if you need them, but reconstruction later sees only S21.

Input: (n_freq, 2, 2) complex S.
Technique: take index [:, 1, 0] in memory, which is S21 from the Touchstone row order.
Output: raw_s21, length n_freq.

Step 8. Sort and put frequencies on a uniform grid

IFFT later requires a constant Δf. The code sorts frequency, then interpolates real and imaginary separately onto a uniform axis. If the file was already uniform (our sample Δf = 40 MHz), interpolation barely changes the numbers, but the flag still records that a uniform grid is guaranteed.

Input: possibly uneven (f, S21).
Technique: interpolate Re(S21) and Im(S21).
Output: uniform_frequencies_hz and S21 on that grid.

Step 9. Magnitude, wrapped phase, unwrap

Wrapped phase jumps by about ±180° when the true delay crosses a branch cut. Unwrap adds or subtracts 360° so the delay curve is continuous. The report stores how many unwrap adjustments happened. This does not change |S21|; it only repairs the angle story.

Step 10. Spike repair

Default method is Hampel. A sample that is far from its local median is replaced. Alternatives are median or local replacement. This is for isolated glitches, not for removing a real resonance.

Input: corrected complex S21.
Technique: Hampel / median / local on the sweep.
Output: corrected_s21 and a count of repaired samples.

Step 11. Optional average and optional reference subtract

If the user supplies repeated sweeps, they are averaged. If a reference measurement is supplied, it is subtracted. On a single teaching .s2p both are usually skipped, and the validation report says Averaging = No, Reference = No.

Step 12. Mild filter on real and imaginary

Default for the .s2p / automation non-BMID path is Savitzky–Golay, window_length = 7, polyorder = 3. The filter is applied to Re and Im separately so phase is not destroyed by filtering |S| alone. NRMSE = ||filtered − before|| / ||before|| tells how much the trace moved.

```
SNR used here (signal vs sweep noise, not image SNR):
smooth |S| with a 5-sample moving average
noise = |S| − smooth
SNR_dB = 10 log10(P_smooth / P_noise)
```

```
dynamic_range_dB = 20 log10(max|S| / min|S|)
```

Step 13. Hamming copy (kept aside)

```
w[k] = 0.54 − 0.46 cos(2π k / (N−1))
S_hamming(f) = S21_filtered(f) · w[k]
```

The Hamming copy reduces IFFT sidelobes later. The GUI still reconstructs from filtered S21(f) by default. The Hamming / Week-3 time arrays are stored in metadata and in the Module 3 handoff file. That gap is documented in the README.

Step 14. Week 3 time-domain (when enabled)

```
Δf = mean consecutive step
Δt = 1 / (N Δf)
t[n] = n Δt
s(t) = IFFT(S_hamming(f))
If a reference exists: subtract the matched Hamming IFFT in time.
If ≥ 2 channels: μ_G[n] = mean over channels; s_clean = s − μ_G
α = max |s|; s ← s / α (global normalize)
```

On sample_touchstone.s2p there is only one channel, so group clutter is skipped and a note is stored. Automation for .s2p turns Week 3 on. Automation for BMID turns it off.

Step 15. General MATLAB stages (non-.s2p)

- Noise filtering: savgol / moving average / Butterworth / median / gaussian, always on Re and Im along frequency.
- Calibration: self or an external reference array.
- Normalization: e.g. divide by the max magnitude.
- Background subtraction: optional, only if do_background_subtraction is true.
- Artifact suppression: hybrid or other methods in artifact_suppression.py.
- Each stage writes a snapshot into stage_outputs so the GUI Stages tab can show before/after.

BMID pass-through sets filter=calibration=normalization=artifact=none, background off, Week 3 off. The output dataset is a copy of the input. That is correct for already-cleaned _adi cubes. Cleaning them again can erase the tumor focus.

Step 16. What Module 2 hands to Module 3

processed_dataset.s_parameters is 2-D (n_freq, n_traces). For .s2p that is (201, 1) containing only S21. For BMID it is (1001, n_antennas). Optional handoff export writes .mat/.csv/.json plus circular Tx/Rx coordinates.

Sample Module 2 output

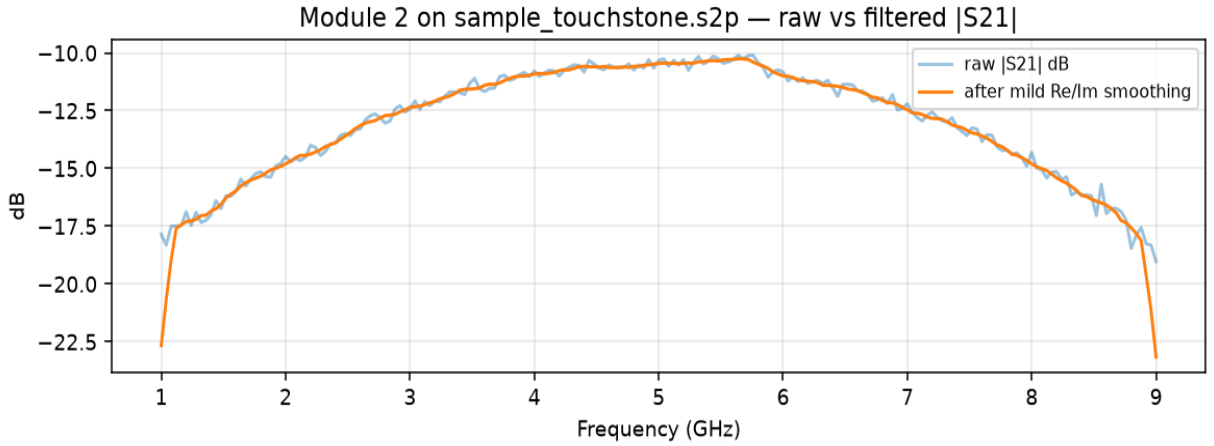


Figure. Raw |S21| (light) versus the same sweep after a mild 7-sample smoother on real and imaginary parts. The overall shape is the same. Sharp jitter is reduced. That is what we want: less noise, same resonance.

How to interpret this figure. If the dark curve had flattened the dip completely, the filter would be too strong. If the two curves were identical down to every wiggle, the filter did almost nothing (that is also fine for an already-clean file). A large NRMSE with a destroyed shape means you should not reconstruct yet — go back to filter settings.

Module 3 — Reconstruction

The purpose now changes again. We stop looking at a line versus frequency. We ask: if each antenna heard a delayed echo, where in the x–y plane would those echoes line up?

Step 17. Read geometry from metadata or defaults

```
x = linspace(x_min, x_max, n_x)
y = linspace(y_min, y_max, n_y)
default n_x = n_y = 64
antenna i: (r cos θ_i, r sin θ_i)
360° ring: θ = linspace(0, 2π, N, endpoint=False)
355° BMID arc: θ = linspace(0, 355°, N, endpoint=True)
optional: angle offset, clockwise, flip x, flip y
optional phase-delay radius: r_eff = 0.97(r − 0.106) + 0.148 m
```

Input: processed traces + ReconstructionConfig.
Technique: infer_reconstruction_config reads antenna_radius_m, wave_speed_m_per_s, FOV spans. Missing values become 0.08 m, 3e8 m/s, ±5 cm. BMID uses 0.18 m and ±6 cm. Phase-delay starts OFF.
Output: antenna coordinates (N × 2 metres) and a pixel grid.

Wave speed c is the conversion from distance to time. Wrong c is like focusing a camera at the wrong distance. Auto Tweak can later try 2.1e8 and 1.8e8 m/s. Batch automation does not, so many files stay comparable.

Step 18. IFFT of every trace

```
s(t) = IFFT(S(f)) along frequency
N_out = n_freq + zero_padding
t[n] = n / (N Δf)
Requirement: Δf nearly constant (relative tolerance 1e-3), at least 2 samples.
```

Input: complex S, shape (n_freq, n_traces).
Technique: NumPy ifft, optional zero padding for finer time bins.
Output: complex time_signals, shape (N_out, n_traces). This is the shared starting point for DAS, DMAS and DMAS-D4.

For sample_touchstone.s2p: N = 201, Δf = 40 MHz, Δt = 1/(201×40e6) ≈ 0.124 ns. Two-way delay from an 8 cm radius antenna to the origin is 2×0.08/3e8 ≈ 0.533 ns ≈ 4 samples. With only one trace, later pairing in DMAS has no partner. That is why the reconstruction pictures below use the 8-antenna synthetic array.

Step 19. Delay-and-Sum (DAS)

```
For every pixel r and every antenna a_i:
d_i = ||a_i − r||
τ_i = 2 d_i / c (two-way time)
n = round(τ_i / Δt)
if n is outside the trace, that antenna adds 0

I_plain(r) = Σ_i |s_i(τ_i)|
CF(r) = |Σ_i s_i|² / (N Σ_i |s_i|²)
I_DAS(r) = I_plain(r) · CF(r)^γ with γ = 0.55
```

γ = 0 is textbook DAS (just add envelopes). γ = 0.55 is the cohort average from the beamformer sweep: it pulls the top spot closer to labeled tumors than γ = 0, while γ = 0.75 is almost the same and γ ≥ 0.80 starts healthy false positives. When antennas disagree, CF is small and that pixel is pushed down.

Input: time traces, antenna positions, grid, c, γ.
Technique: nearest-sample delay, then optional coherence weight.
Output: one real image of shape (n_y, n_x).

Step 20. Delay-Multiply-and-Sum (DMAS)

```
Same delayed samples s_i. For every pair i < j:
p_ij = s_i · conjugate(s_j)
signed_power(z,p) = Re{ sign(z) · (|z|+ε)^p }
p = 0.30 (textbook 0.50), ε = 1e-9
I_DMAS = Σ_{i<j} signed_power(p_ij) · CF^{0}
```

Multiplying two antennas keeps energy that is consistent in both. Random clutter is less likely to survive every pair. That is why DMAS is often sharper than DAS, and also why it is slower.

Step 21. DMAS-D4

```
I_D4 = sign(I_DMAS) · |I_DMAS|^{0.25}
```

Exponent 1/4 is the textbook map and, on the cohort sweep, the average-best. 0.10 through 0.80 were within 0.01 of that peak. 0.00 scattered the spot. 0.85 and above raised healthy false positives. D4 is not a new delay model. It is a contrast map after DMAS.

Step 22. Quality numbers on each image

```
SNR = mean(I) / std(I)
SCR = max(I) / mean(|I − mean(I)|)
CCR = (mean(top 5% pixels) − mean(rest)) / std(rest)
contrast = (max − min) / (max + min)
FWHM_px = average of half-max widths on the peak row and the peak column
```

Step 23. Choose one beamformer

We are still asking “which picture should we trust?” not yet “where is the region?”

```
Quality mode (default when there is no tumor label):
normalize each metric across DAS/DMAS/DMAS-D4 to 0–1
score = 0.35 SNR_n + 0.35 SCR_n + 0.20 contrast_n + 0.10 (1 − time_n)
```

- prefer_dmas_d4: always pick DMAS-D4; still print the other scores.
- force_das / force_dmas / force_dmas_d4: user override.
- tumor_gt: pick the image whose ROI centroid is closest to the labeled (x, y). Used by automation when metadata has a tumor.

Image	SNR	SCR	contrast	quality score
DAS	1.874	6.520	0.744	0.800
DMAS	1.832	5.472	2.134	0.653
DMAS-D4	1.586	3.791	7.911	0.300

On this sample run the winner is **DAS** because it had the highest weighted score. These scores are relative to each other on the same scan. A high score does not mean “tumor present”.

Sample Module 3 output

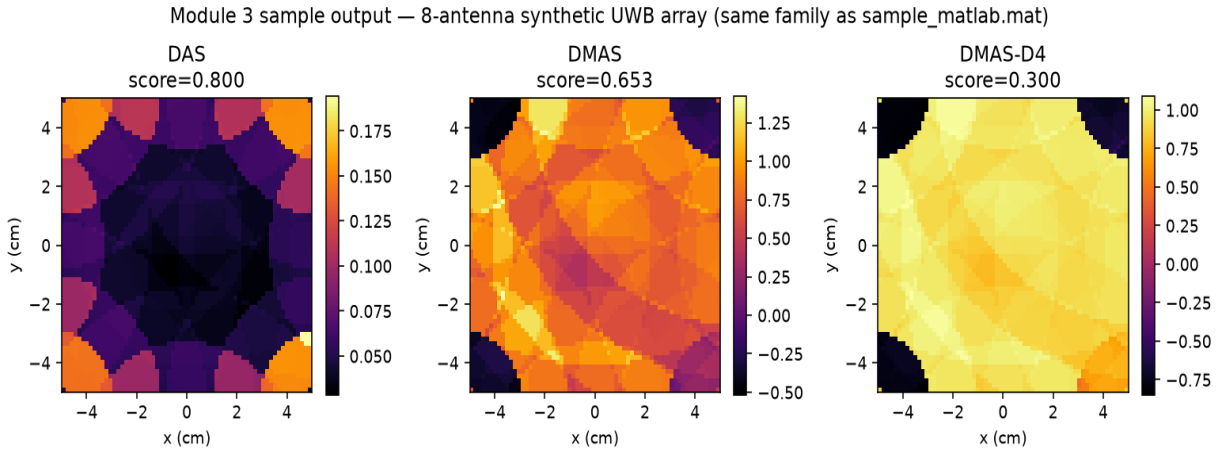


Figure. Three reconstructions of the same 8-antenna synthetic dataset, 64×64 pixels, FOV 10 cm × 10 cm, r = 8 cm, c = 3e8. Color is image intensity (inferno: dark = low, yellow = high).

How to interpret this figure. A useful reconstruction shows a compact brighter patch, not a full bright ring and not a uniformly noisy field. DAS is usually the smoothest. DMAS and DMAS-D4 often look more peaked. If all three images are a circular ring at the antenna radius, the focus is sitting on clutter, not on a compact interior target — later the ring-excess test is designed to catch that. This sample is synthetic teaching data, so the “hotspot” is a made-up resonance, not a labeled tumor.

ROI detection starts

ROI = Region of Interest. The purpose now changes. We are no longer asking “which beamformer is best?” We are asking “where in the selected image is the most relevant / suspicious region?”

Step 24. Image normalization

```
I_n = (I − I_min) / (I_max − I_min + ε)
```

This stretches intensity so the darkest pixel is about 0 and the brightest is about 1. Threshold 0.70 then means “70% of the way from darkest to brightest on this image”, not an absolute physical unit.

Input: selected real image I.
Technique: min–max scale, ε = 1e-12.
Output: I_n in [0, 1].

Step 25. Gaussian smoothing

```
I_s = Gaussian(I_n, σ) default σ = 1.0 pixel

Reconstructed image ↓ Gaussian smoothing ↓ Reduced tiny fluctuations
```

Without this step, a single noisy pixel can become its own “region”. Tight-peak mode may reduce σ further, but the default is 1.0.

Step 26. Thresholding

```
B = 1{ I_s ≥ T } default T = 0.70
```

Meaning: if smoothed intensity ≥ threshold → 1, otherwise → 0. Example (not from the sample image):

```
Intensity: 0.2 0.3 0.8 0.1 0.9 0.7 0.2 0.6 0.3 T = 0.6 gives 0 0 1 0 1 1 0 1 0 The 1s are possible candidate pixels.
```

Then a 3×3 morphological opening removes 1-pixel speckles. That is a tiny clean-up before labeling.

Step 27. Connected-component detection

```
The binary image can contain several separate islands: +----------------------+ | ■■ | | ■■■ | | | | ■■■■■ | | ■■■■■ | | | | ■■ | +----------------------+ Each island is one connected component. Components smaller than min_area = 6 pixels are dropped.
```

If zero components survive, the code raises T by 0.05 (max 0.95) and retries with min_area = 1. If that still fails, it places a small box on the global peak. So ROI detection always returns at least one box.

Step 28. Score each component

```
Score = (0.65 I_peak + 0.35 I_mean) × Compactness × 1/(1 + 10 AreaRatio) × γ_edge
```

I_peak is the strongest smoothed value inside the component. A real hotspot should be bright. I_mean is the average inside the component. Using 35% mean stops the score from worshipping one lucky pixel. Together: 0.65 peak + 0.35 mean.

Step 29. Compactness

```
Compactness = A_component / (A_bbox + ε)

A compact blob fills most of its bounding box: ■■■ ■■■■■ ■■■ A scattered or long smear does not: ■ ■ ■ ■ The second gets a lower compactness and therefore a lower score.
```

Step 30. Area ratio penalty

```
AreaRatio = A_component / (N_x N_y + ε)
size term = 1 / (1 + 10 AreaRatio)
```

If a component covers the whole image, AreaRatio is near 1 and the size term is about 1/11. The system will not simply pick “the biggest bright splash”.

Step 31. Edge penalty

```
γ_edge = 0.65 if the box touches the image border, else 1.0
```

Antenna rings and FOV-edge artifacts often touch the border. They are not deleted, but they lose score.

Step 32. Extra modes and extra spots

- Peak mode (default): score as above.

• Off-center mode: multiply by (0.2 + 0.8 r_norm) × (1 + 1.5 r_norm). r_norm is distance from image centre divided by the corner distance. This fights a false bright centre.

• Tight-peak mode: after finding the peak, keep only a small box (margin about 2 pixels).

• Local-max seeds: a 5×5 maximum filter also proposes peaks above max(0.75 T, 82nd percentile).

• Keep at most 4 spots. A weaker spot is kept only if its score ≥ 0.40 × best score (automation uses 0.28) and it is at least 6 pixels from a stronger spot.

• If a labeled tumor (x, y) exists, the nearest peak within 3 cm is added even if it lost the spacing test. Labels do not delete other spots.

Step 33. Best ROI is selected

```
On this sample run the scored components were: Component 1: score=0.357, area=93 px,
compactness=0.775, peak=0.929, mean=0.757, area_ratio=0.0227, edge_γ=0.65 Component 2:
score=0.310, area=94 px, compactness=0.777, peak=0.758, mean=0.748, area_ratio=0.0229, edge_γ=0.65
Component 3: score=0.304, area=93 px, compactness=0.769, peak=0.751, mean=0.740,
area_ratio=0.0227, edge_γ=0.65
```

Component 1 has the highest score (0.357), so it becomes the primary ROI. If several spots stay, the primary is still the top score. Localization against a label later uses the spot closest to the label, which can be a different spot.

```
Illustration of the idea (not these exact sample scores): Component A → 0.31 Component B → 0.67
Component C → 0.44 → B is selected.
```

Step 34. High-resolution ROI reconstruction

```
Full image (64×64) ↓ Selected bounding box ↓ Convert box pixels back to metres ↓ Rebuild DAS or
DMAS or DMAS-D4 only in that rectangle ↓ grid size = clip(box_size × 4, min 96, max 256)
```

Input: processed S, chosen algorithm, coarse box, full FOV.
Technique: same delays, denser pixels, smaller span.
Output: ROIRefinement: refined image, local x/y spans, new grid shape. This is zoom-by-physics, not a Photoshop zoom.

Sample ROI output

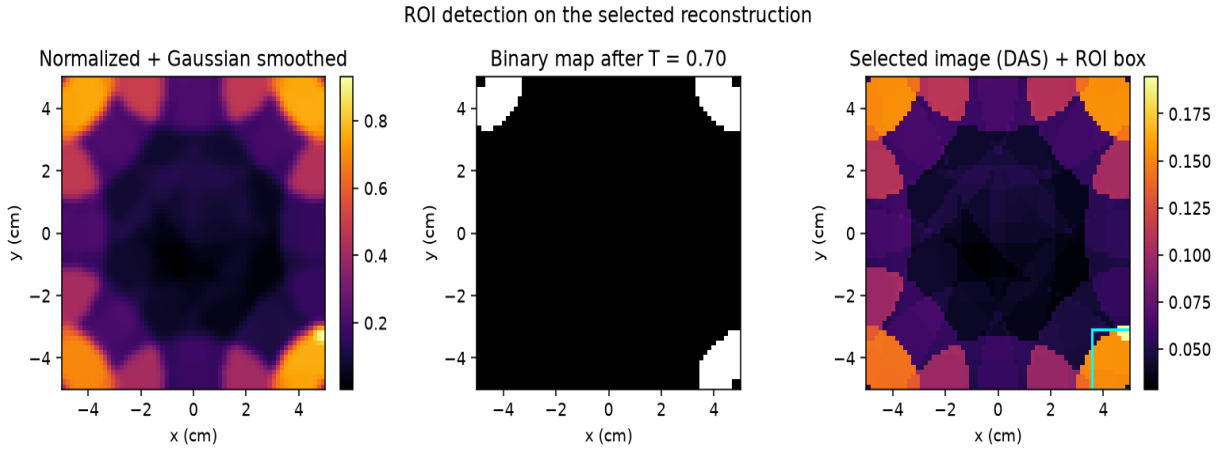


Figure. Left: normalized and Gaussian-smoothed selected image. Middle: binary map at T = 0.70 after opening. Right: original selected reconstruction with the winning box in cyan.

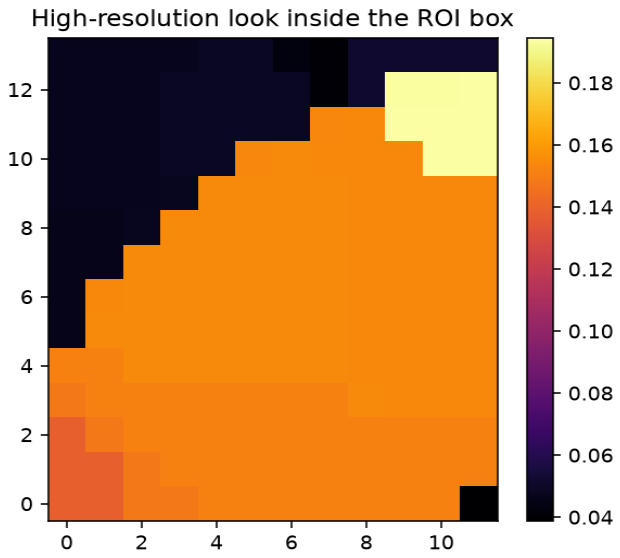


Figure. Crop / refined look inside that box.

How to interpret these figures. The middle panel should show one or a few white islands, not a white snow field. A snow field means T is too low or the image has no compact peak. The cyan box should sit on the island you would point to by eye. The refined crop on this synthetic sample shows a local intensity gradient and some diagonal texture. It is not a perfectly round clinical “mass”. That is expected for a teaching resonance, and it is exactly why later suspicion and ring tests exist: a box can be drawn around clutter. Drawing a box is not a diagnosis.

Step 35. ROI characterization (describe the box)

The purpose changes again. We are no longer choosing a box. We are measuring it.

- Centroid (cm): pixel centre mapped through the FOV. This is the (x, y) used for localization error.
- Area in pixels and in cm² using pixel size = FOV / (N–1).
- Equivalent diameter = 2 sqrt(area_cm² / π).
- Bounding-box width, height, aspect ratio.
- Peak intensity and mean intensity inside the mask; peak/mean.
- Local SCR = SCR computed only inside the box.
- FWHM in cm = FWHM_px × average pixel size.
- Compactness = area_px / (bbox_w × bbox_h).
- Eccentricity proxy = (aspect–1)/(aspect+1).
- Radial offset = distance of centroid from the origin, in cm.

Input: selected image + ROI box + FOV.
Technique: characterize_tumor_region in quality/characterization.py.
Output: a list of centimetre and intensity descriptors. Still not YES/NO.

Step 36. Tumor-candidate scoring (a new question)

This is separate from ROI selection. A region can win the ROI contest and still fail the tumor-like test. The implemented suspicion is not a single peak/mean fraction. It is a blend of four 0–1 features:

```
local_contrast = I_peak_inside / (|mean_outside| + ε)
ring_excess = I_peak / (median of other pixels on the same radius + ε)
size_score = 1 / (1 + (AreaRatio / 0.08)²)
contrast_score = tanh((local_contrast − 1) / 2)
ring_score = tanh((ring_excess − 1.15) / 0.35)
location_score = 0.45 + 0.55 r_norm
Suspicion = clip(0.38 contrast_score + 0.22 size_score + 0.15 location_score + 0.25 ring_score, 0,
1)
```

Why ring_excess exists: a focus on the antenna ring is bright at one angle, but the rest of that radius is also bright. A compact interior target is bright while the rest of that radius is dark. location_score has a 0.45 floor so a tumor near the origin is not zeroed.

Step 37. Candidate confidence and penalties

```
Conf = clip(Suspicion × p_edge × p_area × p_ring, 0, 1)
p_edge = 0.85 if the box touches the border, else 1
p_area = 0.65 if AreaRatio > 0.35, else 1
p_ring = 0.55 if ring_excess < 1.20, else 1
```

If a labeled tumor is within 2 cm, Suspicion is increased by 0.35 (cap 1). Within 3 cm, +0.20. That is a research aid when GT exists. It is not used on unlabeled .s2p files.

Step 38. Final YES / NO

```
YES if Conf ≥ 0.45
also YES if GT distance ≤ 3 cm and local_contrast ≥ 1.05 and AreaRatio ≤ 0.40 (Conf at least 0.55)
forced NO if ring_excess < 1.40 and (no GT or GT > 3 cm)
```

Reasons the code stores: localized to labeled tumor; compact hotspot contrast is tumor-like; peak sits on a rotationally symmetric clutter ring; large edge-touching clutter; or weak evidence.

```
Numbers from this sample reconstruction (no GT): peak=0.1945, local_contrast=2.907, r_norm=0.841,
area=120 px, area_ratio=0.029, ring_excess=1.354 Suspicion=0.744, Confidence=0.632, decision=NO
```

How to interpret this decision. This sample has no laboratory tumor label, so GT boosts are off. The decision is **NO** with confidence 0.63 versus the 0.45 gate. ring_excess = 1.35. If that number is below 1.40 the code treats the peak as a symmetric ring and forces NO. A NO here means “this teaching image does not look like a compact interior target under the written rules”. It does not mean the file is broken, and it does not mean a patient is healthy.

If several spots exist, each is classified. Combined YES if any spot is YES. Combined confidence is the maximum spot confidence. The sentence becomes “K of N spatially separated hotspots look tumor-like.”

Step 39. Overall reconstruction confidence (another 0–1 score)

This is not the same as candidate confidence. It blends image quality, detection, how clearly one beamformer won, localization, and focus sharpness.

```
image_q = 0.45(1 − e^{−SCR/8}) + 0.30 contrast + 0.25(1 − e^{−CCR/4})
detection = candidate confidence
margin = clip(0.55 + 0.9 tanh(2 (score_sel − score_2nd)))
localization = e^{−dist_cm/2.5} if GT else 0.55 compactness + 0.45 e^{−|diam−2|/2.5}
focus = 0.6 e^{−max(FWHM−1.5,0)/3} + 0.4 e^{−max(area_cm²−4,0)/6}
overall = 0.22 image_q + 0.28 detection + 0.15 margin + 0.22 localization + 0.13 focus
label: high ≥ 0.75, medium ≥ 0.45, else low
```

Step 40. Localization error when a label exists

```
error_cm = 100 × hypot(centroid_x − tum_x, centroid_y − tum_y)
```

Balanced automation warns if this is above 3 cm. Strict uses 1.5 cm. Lenient uses 5 cm. sample_touchstone.s2p has no tumor label, so this number is n/a.

Module extras that sit beside the main chain

Step 41. Auto Tweak (one scan, many geometries)

Input: current ReconstructionConfig plus optional tumor (x, y).
Technique: score the manual baseline; DAS-only sweep of angles {0,90,180,270}, flips, CW/CCW, 360/355°, phase-delay on/off, ROI modes peak/off-center/tight; then full DAS/DMAS/DMAS-D4 on the top 8.
With GT: score = 1/(distance + 1e-4). Apply a change only if error improves by at least 0.5 mm.
Without GT: prefer contrasty, off-centre, compact ROIs.
Output: best geometry + beamformer, or the original settings if nothing clearly won.

Step 42. Tumor taxonomy (labels, not the image)

If metadata says there is a tumor: size small ≤ 2 cm, medium ≤ 4 cm, else large. Severity is a placeholder from size and BIRADS (BIRADS ≥ 4 can raise the bucket). Quadrant from (x, y), with a 1 cm disk called central. Shape is the metadata string. Healthy scans are labeled healthy. These are filing labels. The learned reading of tumour, size, diameter and criticality is Steps 45–60.

Step 43. Automation wraps the same chain

Discover files → pair metadata to measurements → pick scans (diverse / all / …) → for each target run Steps 1–40 with a named profile (strict / balanced / lenient) → write latest_auto_analysis.md, JSON, and PNG figures → optionally open the showcase case in the GUI. Hard fail: load error, NaN/Inf image, energy too small, ROI area too small. Soft warning: localization, SCR, SNR, confidence, or GT-tumor-but-not-candidate.

Step 44. What the GUI shows at each module

- Acquisition: file picker, BMID scan list (HEALTHY / TUMOR text), dataset summary, status log, Run Auto Validation.
- Preprocessing: preset, stage plots (raw / filtered / windowed / time), quality numbers, export handoff.
- Reconstruction: radius, FOV, c, beamformer mode, ROI mode, Run, three images, ROI overlay, refine tab, Auto Tweak, session report under results/.

Learned tumour classification, step by step

The chain above ends at a rule-based YES or NO. The learned classifier is a separate module. Each step below is the same kind of record: the input, the formula, and the number that formula produced on the fair UM-BMID test.

Module — Tumour classification (the learned reading)

Steps 1-44 turn one file into a picture and a rule-based YES or NO (confidence at least 0.45). That rule does not learn from other phantoms. On the fair UM-BMID test it scores about 50/100, which is a coin flip. This module is the separate learned reading. It asks three questions of phantoms it has never seen: is there a tumour, how big is it, and where is it. Official first-scan score: **71 +/- 2/100**. Official follow-up score: **97 +/- 0/100**. Neither number is a medical diagnosis.

Step 45. Two situations, two subtractions

Input: one UM-BMID scan S(f, antenna), shape (1001, n_antennas), f = linspace(1 GHz, 8 GHz, 1001), plus the empty-chamber scan of that sitting.
First visit has no earlier healthy scan of the same breast. Only the empty chamber is removed.
Follow-up also subtracts earlier healthy scans of the same phantom and the same sitting.

```
S_cal = S - S_empty
S_rot = S_cal - mean_over_antennas(S_cal)
S_follow = S_cal - mean(S_healthy_1 ... S_healthy_k)
```

The rotational mean is the part that looks the same from every antenna: skin, fat shell, chamber. A compact inclusion does not look the same from every angle, so it survives in S_rot. S_follow is stronger when a previous healthy scan exists, because that scan carries the fibroglandular tissue as well as the shell. That is why the follow-up score is 97 +/- 0/100 and the first-scan score is 71 +/- 2/100.

Step 46. Breast time window

The frequency samples are turned into time with an IFFT. The delay of sample n is t_n = n / (N * df). Echoes before 0.71 ns are the antenna. Echoes after 5.71 ns are past the breast. This is the same window used by the published logistic-regression baseline on this dataset.

```
dt = 1 / (N * mean(diff(f)))
keep samples with 0.71e-9 <= t <= 5.71e-9
about 36 time samples on the 1-8 GHz, 1001-point grid
```

Step 47. Signal features (prefix sig_ and rsig_)

Input: S_cal for sig_, and S_rot for rsig_. No tumour coordinate is used.
Output: one number per formula below. The forest and the logistic model see only these numbers, not the raw matrix.

```
cal_to_raw = log10( sum|S_cal|^2 / sum|S|^2 )
rot_residual = sum|S - mean_antenna|^2 / sum|S|^2
For each 1 GHz band from 1-2 through 7-8:
band_log = log10(mean power)
band_cv = std(power per antenna) / mean(power)
band_rot = rotational energy in that band / band energy
```

```
For each kept time sample k:
w_log = log10(mean over antennas of |s(t_k)|^2)
w_cv = std / mean of that time sample
Angular harmonics h1..h4 = |FFT of antenna power|[k] / |FFT|[0]
ant_cv = std(antenna energy) / mean(antenna energy)
ant_max_over_median = max(antenna energy) / median(antenna energy)
```

sig_ is the empty-chamber-cleaned scan. rsig_ is the same list after the rotational mean is removed. The breast-only candidate used later keeps just the w_ numbers (the time samples inside 0.71-5.71 ns) and drops the frequency-band numbers, because those bands still contain the antenna reflection.

Step 48. Image features (prefix img_ and imgl_)

Coherent delay-and-sum is run on S_rot. The grid is 64 by 64 over +/- 6 cm. Wave speed and antenna layout are the fitted values for that generation (gen1: 2.7e8 m/s, gen3: 2.1e8 m/s). imgl_ uses only 1-4 GHz, which was the band that localised gen3 best. DMAS-D4 features are stored but are not allowed into the model search: they were built before a baseband fix.

```
peak_over_mean = max(image) / mean(image)
ccr, FWHM_cm, entropy, top-1-percent / median
peak (x, y) of a lightly smoothed image, and its radius
From the brightest blind ROI (no ground truth):
equivalent diameter = 2 * sqrt(area / pi)
aspect, compactness = area / bounding box
eccentricity = (aspect - 1) / (aspect + 1)
local contrast, ring excess, suspicion, rule confidence
The rule confidence is a feature. It is not the classification answer.
```

Step 49. Physics summaries (the side check)

These 15 numbers describe the rotational residual with formulas, instead of handing every time sample to the forest. They were tested after the official model. They did not replace it.

```
Breast-window matrix B (time by antenna), singular values s_i
L_i = s_i^2 / sum(s_j^2)
effective rank = 1 / sum(L_i^2)
On 2-6 GHz, p(f) = mean over antennas of |S_rot|
log10 p(f) = a + b * log10(f / 1 GHz) (ordinary least squares)
A small sphere in the Born approximation has b near 2.
Excess kurtosis of the energy-weighted delay
Angular resultant R = |sum p_k exp(i theta_k)| / sum p_k
Radar range of the brightest antenna:
d = R_antenna - c * t_peak / 2
```

Measured on the same fair first-visit scans. Signal features alone in this narrower test: 69/100. Signal plus these formulas: 68/100. Formulas alone: 51/100. The official nested first-scan score remains 71 +/- 2/100, because that test is allowed the image features and the extra training scans. On a first visit the tumour does not move these summaries more than healthy fibroglandular tissue does.

Step 50. The fair test set

Why. Tumour scans and healthy scans were often taken in different sittings, and the hardware drifts. A model can look clever by recognising the sitting. **Rule.** Keep only sittings that contain both classes. Inside each sitting, draw the same number of tumour scans and healthy scans (the smaller of the two counts, random seed 0). **Resulting test:** 268 first-scan scans from 27 phantoms (134 tumour). Follow-up: 246 scans from 24 phantoms. Generation 2 is left out of the follow-up test: its tumour scans and healthy scans were different sittings, so the sitting number alone gives the answer.

A second model sees only antenna radius, sitting number and generation. On the first-scan test it scores 47/100. On the follow-up test it scores 49/100. Both must stay near 50. If they do not, the sitting is leaking the label.

Step 51. Phantom-grouped nested testing

Input: the fair scans, grouped by phantom name.
Technique: StratifiedGroupKFold. A phantom is never in both the training fold and the test fold. Outer loop: 5 folds. Inner loop, on the training phantoms only: 4 folds. Repeated 3 times with different seeds.
What is chosen inside the inner loop, and nowhere else: which feature family, which model, which training pool, and the

alarm cut-off.

```
Feature families:
signal = sig_
signal_rot = rsig_
breast = time-window samples only (sig_w and rsig_w)
image = img_ and imgl_
all = signal + signal_rot + image
Models: logistic, or a forest of 400 trees
Pools: fair scans only, or fair scans plus every other scan of phantoms that are not in the test
fold
The test fold is scored once, with the choice the inner loop made. The reported +/- is the spread
of that score across the 3 repeats.
```

What the inner loop picked most often. First scan: all|random_forest|fair_plus_other. Follow-up: signal_rot|logistic|fair. The saved models in results/classification/ are those choices, fit afterwards on all training phantoms. The scores above are the nested test, not the fit on every phantom.

Step 52. The two learners, written out

```
Logistic, used for the official follow-up:
z_j = (x_j - mean_j) / std_j (means and stds from the training phantoms)
p = 1 / (1 + exp(-(b0 + sum b_j z_j)))
penalty: inverse strength C = 0.5, class_weight = balanced
balanced means each class is given the same total weight, so a larger class cannot dominate

Random forest, used for the official first scan:
400 trees, a leaf must hold at least 3 scans
class_weight = balanced, random_state = 0
the score is the fraction of trees that vote tumour

Diameter uses ridge regression (penalty alpha = 3) or a forest regressor (400 trees, min leaf 3,
max_features 0.33). Size class uses the same logistic or forest, with three labels.
```

Step 53. The yes/no cut-off

The model outputs a probability p. A scan is called tumour when p is at least t. t = 0.5 is reported, but it is not the official alarm. The official t is chosen on the training phantoms only, at the point where the true-positive rate minus the false-positive rate is largest (the Youden point).

```
t = argmax over thresholds of (TPR - FPR)
TPR = tumours found / all tumours
FPR = healthy scans called tumour / all healthy scans
sensitivity = TPR, specificity = 1 - FPR
```

Saved first-scan cut-off: 0.507. At that cut the nested test finds 66% of tumours and clears 65% of healthy scans. At the default 0.5 it finds 74% and clears 57% (more tumours, more false alarms). Saved follow-up cut-off: 0.663. It finds 88% and clears 95%.

Step 54. How a follow-up reference is chosen

Inside one phantom and one sitting, list the healthy scans. For every scan in that sitting, tumour or healthy, the other healthy scans are the allowed references. Let m be the smallest number of other healthy scans any one of those scans has. Use k = min(3, m). If m is 0, the sitting is dropped. The k scans are drawn with a random generator seeded by the scan id, the same way for both classes. The reference is their average. A tumour is never given a cleaner baseline than the healthy scans, so k cannot give the label away.

Measured. Follow-up score 97 +/- 0/100 (older phantoms 86, newest 99, tumours of 2 cm or smaller 97). The change in the scan grows with tumour diameter on the newest phantoms, which is the check that the subtracted signal is the tumour and not a processing accident. A 1 cm tumour changes the scan about as much as repeating a healthy scan.

Step 55. Size, diameter, type and criticality

```
size = small if d <= 2 cm
medium if d <= 4 cm
large otherwise
Balanced accuracy = average of the three class recalls.
```

```
Chance for three equal classes = 1/3.
Diameter error = mean |predicted cm - true cm|

Type would be spherical versus spiculated. Only 6 of 134 fair tumour scans are spiculated, all in
the older phantoms, so no shape classifier is reported.

Criticality is a research index, not clinical BI-RADS (the metadata BI-RADS number is breast
density):
C = (d / 2 cm)^3
C = 1.5 * C if the shape string contains spiculated
low if C <= 1, moderate if C <= 8, high if C > 8
A 2 cm sphere is low (C = 1). A 2 cm spiculated inclusion is moderate (C = 1.5). A 4 cm sphere is
moderate (C = 8). A 6 cm sphere is high (C = 27).
```

First-scan size class: 64 +/- 3/100 balanced accuracy (recording-details-only 57). First-scan diameter error: 0.7 cm (older 1.1 cm, newest 0.6 cm). Follow-up size class: 56 +/- 1/100, and 90% correct on the newest phantoms. Follow-up diameter error: 0.4 cm, newest phantoms 0.2 cm. The volume-index classifier on the first-scan fair tumours scores 71 +/- 3/100 balanced accuracy (chance 33). Adding the physics summaries leaves it at 71/100.

Step 56. Where the tumour is

A learned position model is trained during the nested test and then not saved. Inside one sitting the tumour sits in the same place, so a model that can tell sittings apart places the tumour without reading the image. The honest localiser never sees a label or a sitting: it is the brightest spot of the 1-4 GHz coherent image (imgl_peak_x_cm, imgl_peak_y_cm).

```
miss_cm = hypot(x_peak - x_true, y_peak - y_true)
radar alternative, not used as the official localiser:
d = R_antenna - c * t_peak / 2
(x, y) = d * (cos theta, sin theta) at the brightest residual antenna
```

On the fair tumour scans the image spot misses by 1.76 cm on average (65% within 2 cm). The nested trained position model misses by 2.04 cm, which is worse, and it is the one that can borrow the sitting. The closed-form radar range, on the same 133 tumour scans, misses by 2.57 cm (44% within 2 cm). The image spot on those scans misses by 1.76 cm (65% within 2 cm). On gen3 the radar range misses by 2.19 cm (48% within 2 cm). Gen1 images still lose to guessing the centre: the dense inner tissue is off-centre and brighter than the tumour. Gen3 is the generation where the image spot is the result to quote. The gen3 antenna layout was fitted using tumour positions, and the held-out half is the number that should be quoted for that layout.

Step 57. What happens for one new scan

Input: the scan, its empty-chamber scan (or the follow-up healthy average), antenna radius, generation.
Steps inside predict.py: S_cal, S_rot, signal features, coherent DAS, low-band DAS, image features, physics summaries.
Then: the saved forest (first scan) or the saved logistic model (follow-up) turns that vector into p. Compare p with the saved Youden cut-off. Size class and diameter come from their own saved models. Criticality is computed from the predicted diameter with the spherical factor, unless a shape string is supplied.
Output: tumour probability, yes/no, size class, diameter in cm, criticality, and the image-spot (x, y). The image spot is marked reliable only for gen3.

Step 58. How the 71 and the 69 fit together

71/100 is 71 +/- 2: nested test, feature family chosen inside the training phantoms, and that choice was signal plus image, random forest, trained with the extra scans. 69/100 is the later side check: the forest was locked to signal features and to the fair training scans only, before the formulas were added. Adding the formulas made 68. Same phantoms, smaller menu of inputs. The 69 did not replace the 71. Oldest-phantom first scans are the weak part (67/100). Newest-phantom first scans are 72/100. Follow-up on those same newest phantoms is 99/100.

Step 59. What was tried and did not raise the first scan

Pasting a tumour echo measured on a training breast onto other training breasts. The fair test breasts were never touched. The forest score fell.

Subtracting an average fat shell built from other breasts. That removed the tumour as well as the shell, and the first visit stayed with the empty chamber only.

Removing the strongest shared patterns (SVD) and cutting the early skin echo. The shared-pattern removal took the tumour with it. The time cut changed nothing.

The physics summaries in Step 49. They describe the scan, and they do not separate tumour from healthy on a first visit.

A convolutional network on the breast-window sinogram, which is the input behind the published 78 ROC figure, and a PCA-then-logistic classifier on that same sinogram, which is the simulation papers' step. On this fair phantom-held-out test the network scored 50/100 and the PCA logistic 45/100. The network could fit the breasts it trained on (about 71/100 on those breasts) and the score fell to about 58/100 on a held-out training phantom, then to 50/100 on the fair test. Neither replaced the forest.

Step 60. Scores next to the step that produces them

```
Step 45-48 features, nested, first scan: 71 +/- 2/100 (Step 51-53).
Step 45 follow-up subtraction, nested: 97 +/- 0/100 (Step 54).
Step 38 rule checker, same first-scan scans, no training: 50/100.
Step 53 cut-off, first scan: find 66%, clear 65%.
Step 55 size, first scan / follow-up: 64 +/- 3 / 56 +/- 1 balanced accuracy out of 100.
Step 55 diameter, first scan / follow-up: 0.7 cm / 0.4 cm.
Step 56 image spot: 1.8 cm average miss.
Sinogram network / PCA logistic, same fair first-scan test: 50/100 / 45/100. Not adopted.
Files: results/classification/final_summary.json holds every nested number.
results/reports/classification/guide_progress.pdf is the short sheet for a project guide.
```

Two complete walks

Walk A — sample_touchstone.s2p (what you can run immediately)

- Input form: 2-port RI Touchstone, 201 points, 1–9 GHz, R = 50 Ω.
- Module 1 output: S shape (201, 2, 2); comments set r = 8 cm, c = 3e8, FOV ±5 cm.
- Module 2 output: S21 only, shape (201, 1); mild Savitzky–Golay; Week 3 IFFT; clutter skipped.
- Module 3 limitation: one trace, so DMAS pairing is empty. The image is a poor teaching example. Use the MATLAB 8-antenna file to see beamformers.
- ROI still runs. Tumor-candidate is usually NO / Unknown because there is no GT and a single-trace image is not a compact interior focus.
- Automation profile lenient can still PASS load/energy gates. That PASS means “the pipeline finished cleanly”, not “a tumor was found”.

Walk B — UM-BMID when the cube is on disk

- Input form: fd_data_s21_adi.mat cube + md_list_s21_adi.mat. Choose scan k.
- Module 1 output: S shape (1001, n_ant), tumor (x, y, d) if labeled.
- Module 2 output: same S (pass-through).
- Module 3: r = 18 cm, FOV 12×12 cm, 64×64, three beamformers, tumor_gt selection if labeled.
- ROI + refine + suspicion. Localization error compared with 3 cm (balanced).
- Taxonomy string such as small/moderate/upper-left is copied from metadata. The learned scores are Steps 45–60.

Cheat-sheet of every decision number

Decision	Value	Step
Supported files	.mat .s1p .s2p .s4p .s8p	1
BMID frequencies	1001 from 1–8 GHz	4
GHz if max(f)<1000	× 1e9	3
Default r, c, FOV	8 cm, 3e8, ±5 cm	17
BMID r, FOV	18 cm, ±6 cm	4, 17
IFFT Δf tolerance	rtol 1e-3	18
DAS γ	0.55	19
DMAS p, CF γ, ε	0.30, 0, 1e-9	20
D4 exponent	0.25	21

Decision	Value	Step
Quality weights	0.35 / 0.35 / 0.20 / 0.10	23
Savgol window, order	7, 3	12
Hamming	0.54 − 0.46 cos(2πk/(N−1))	13
ROI T / σ / min area	0.70 / 1.0 / 6 px	24–27
Opening kernel	3×3 ones	26
Score mix	0.65 peak + 0.35 mean	28
Edge γ	0.65 if touches border	31
Keep weaker spot	0.40×best (auto 0.28)	32
Spot gap / max spots	6 px / 4	32
Refine grid	×4, clip 96–256	34
Suspicion weights	0.38 / 0.22 / 0.15 / 0.25	36
Candidate gate	Conf ≥ 0.45	38
Ring veto	ring_excess < 1.40	38
Conf penalties	0.85 edge, 0.65 big, 0.55 ring	37
GT boost	+0.35 ≤2 cm, +0.20 ≤3 cm	37
Overall conf blend	0.22/0.28/0.15/0.22/0.13	39
High / medium label	0.75 / 0.45	39
Balanced loc. gate	3.0 cm	40
Auto Tweak improve	0.5 mm vs baseline	41
Size classes	≤2 / ≤4 cm	42
Phase-delay r	0.97(r−0.106)+0.148	17
Empty chamber	S_cal = S − S_empty	45
Rotational mean	S_rot = S_cal − mean over antennas	45
Breast window	0.71–5.71 ns	46
Nested folds	outer 5, inner 4, ×3	51
Logistic C	0.5, class-balanced	52
Forest	400 trees, leaf ≥ 3	52
Alarm cut-off	argmax(TPR − FPR)	53
Follow-up count	k = min(3, fewest other healthy)	54
Criticality C	(d/2 cm)^3, ×1.5 if spiculated	55
Official localiser	imgl peak, 1–4 GHz	56

If a result looks wrong, walk backwards

- Did Module 1 load a measurement file, not metadata-only?
- For BMID, was a scan index chosen?
- Was BMID pass-through used on already-clean _adi data?
- Are r and c the physical values for that array?
- Do the three images look like a ring? Try Auto Tweak or off-center ROI before inventing a new beamformer.
- A PASS in automation is a quality-gate result. A YES is a tumor-candidate rule. A probability from Steps 45–60 is a research score. None of these is a diagnosis.

That is the project from A to Z: a file becomes a complex matrix, the matrix is cleaned or left alone, delays turn it into three pictures, one picture is chosen, a box is scored, the box is measured, a written rule says YES or NO, and a model trained on other phantoms returns a probability, a size and a criticality index. Every arrow has an input, a technique, and an output. A person still has to read the picture.

Tumour classification status

Official first-scan score: **71/100**. This is the nested test on UM-BMID (signal plus image, extra training scans from phantoms that are not in the test fold, cut-off chosen on training phantoms only). It has not been replaced.

Official follow-up score: **97/100**. The scan is compared with earlier healthy scans of the same breast from the same sitting. Tumour and healthy cases are given the same number of those scans, at most 3, so the count cannot give the answer away.

A later side check scored about **69/100**. That check used only the signal features and only the fair training scans. Adding scattering-spectrum and frequency-slope formulas did not raise it. It is recorded as a side check, not as a new official score.

A convolutional network on the breast-window sinogram, and a PCA-then-logistic classifier on that same sinogram, were then scored on the fair phantom-held-out test. They scored **50/100** and **45/100**. The network could fit the breasts it trained on and did not carry that fit to a new phantom. Neither replaced the official first-scan model.

Two write-ups live in results/reports/classification/: **how_classification_works.pdf** (process, cut-offs and formulas, in plain language) and **guide_progress.pdf** (numbers and progress for a project guide). Machine-readable scores stay in results/classification/final_summary.json.